Structure-Mapping Memory Grounds Language Models in Curated Ontologies

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Abstract

Retrieval-augmented generation (RAG) grounds large language models in external text, yet vector and knowledge-graph retrievers index surface similarity and entity adjacency, discarding two things experts reason with: the subsumption (is-a) hierarchy and higher-order relations. We present Structure-Mapping Agentic Memory (SMA), a retrieval memory that encodes each observation as a functor over a subject and retrieves by analogical structure-mapping against a curated, expert-maintained domain ontology mounted as its ascension lattice. A single universal loader ingests a broad class of open OBO/OWL ontologies (the is-a hierarchy plus `someValuesFrom`/`allValuesFrom` and OBO relationships, with no OWL reasoning); with it we mount eleven ontologies spanning seven knowledge areas (≈594,000 concepts) and evaluate a memory-swap benchmark on five, all served by one shared retrieval core plus thin per-domain data adapters. With the retriever as the only moving part, SMA outperforms a strong RAG and knowledge-graph baseline suite on the rare, long-tail slice in medicine (**+33**), genomics (**+16**) and finance (**+17** percentage points top-5) and on patent classification (legal **+6**, all-query slice, as CPC has no rare split); all four are Holm-significant across domains and survive a conservative selection correction for the data-chosen best baseline. The same trend appears in cyber (**+7**) but is weaker and does not survive that correction. Ablations attribute this measured advantage to bounded is-a subsumption together with information-content weighting; higher-order relational structure is implemented and is decisive on a synthetic structure-only control, yet adds no measurable top-5 gain on the present real-domain benchmarks. Against a hand-built ontology information-content tool, SMA draws even. End to end, holding the language model and prompt fixed and swapping only the memory, the SMA-grounded agent is more accurate, more selective, and more structurally attributable than dense RAG in a medicine case study, separating known from unknown entities (grounding AUROC **0.79** versus vector RAG's near-chance **0.55**) so that it abstains and flags novelty. We delimit the thesis honestly: the advantage is specific to ontology structure and vanishes on flat-tabular prediction.

Keywords: structure mapping, analogical retrieval, ontology, retrieval-augmented generation, abstention, novelty detection

1 Introduction

Large language models are fluent generalists but unreliable specialists. They confabulate confidently on the rare, the long-tail, and the high-stakes cases, which is precisely where expertise is required and where a wrong answer is most costly [1, 2]. The dominant remedy, retrieval-augmented generation (RAG) [3], conditions the model on documents fetched by embedding similarity [4] or lexical overlap [5]. When the answer is frequent and lexically similar to the query, this works well; on the cases that define expertise it fails, because vector retrieval averages rarity away and indexes surface form, so the decisive rare term is washed out. Knowledge-graph and graph-RAG retrievers [6, 7] add typed entity edges, yet they traverse them by path adjacency and still discard the two things experts actually reason with: the *subsumption lattice* (what generalizes to what) and the *higher-order relations* (relations over relations). What remains is a retriever that cannot tell a known case from an unknown one, and therefore cannot know when to abstain.

The gap is representational, not a matter of scale. Expertise is organized as structure: a clinician reasons over an is-a hierarchy of phenotypes and diseases, and an analyst reasons over a typology of techniques and the relations between them. How humans retrieve and reason over such structure has been formalized by decades of cognitive science. Structure-mapping theory [8] holds that analogues are aligned by the systematicity of their relational structure rather than by surface features; building on it, the structure-mapping engine (SME) computes consistent mappings [9], and the MAC/FAC model retrieves candidate analogues tractably from a large memory before aligning them [10, 11]. These mechanisms have never been operationalized as a retrieval memory for grounding language models.

We do so here. SMA treats a curated, open domain ontology as retrieval geometry. Each observation becomes a functor over a subject; the ontology's is-a tree becomes an ascension lattice along which specific terms match general ones (with penalty ρ^{dist}); its typed relations become higher-order statements; and an information-content scorer ($-\log_2 p$) makes the rare-but-decisive term dominate by construction. Our thesis is that *a structure-mapping memory grounded in a curated (expert-maintained) domain ontology makes a generalist LLM a more reliable and attributable specialist*, beating vector RAG and knowledge graphs on rare, cross-vocabulary, high-stakes reasoning while supplying the provenance, calibrated abstention, and novelty detection that those retrievers structurally lack. We are careful about *which* part of the mechanism is load-bearing: on the real-domain benchmarks the measured advantage is carried by bounded is-a subsumption together with information-content weighting, whereas the higher-order relational machinery, though implemented and decisive on

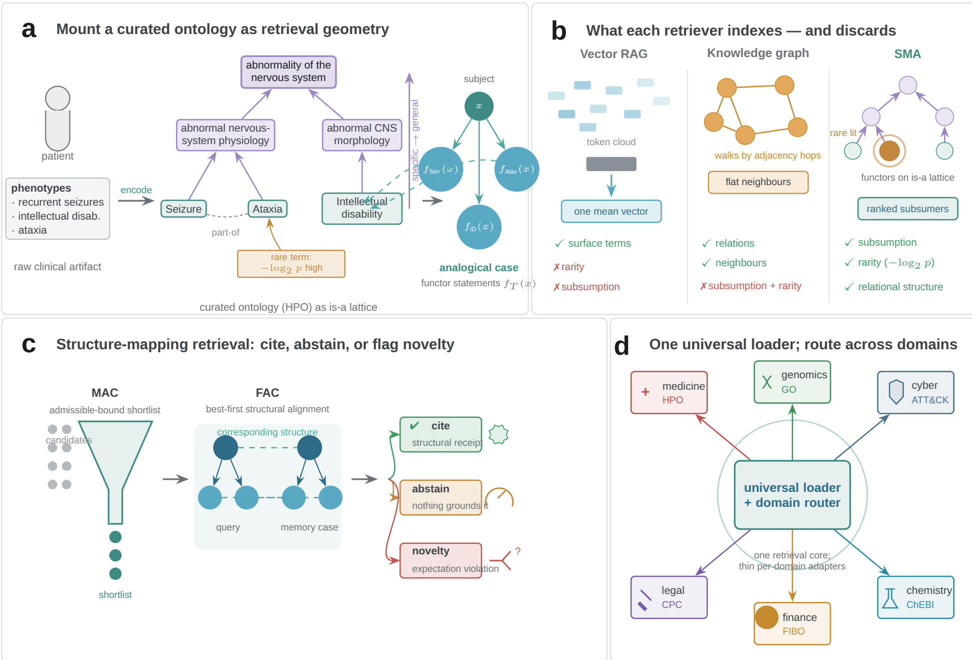

Fig. 1 **The universal-adapter system.** (a) Any open ontology is mounted as retrieval geometry: an artifact is encoded as functors over a subject, the is-a tree becomes the ascension lattice, and typed relations become higher-order statements. (b) What each retriever indexes and discards: vector RAG (token similarity, discards rarity), knowledge graph (entity adjacency, no subsumption), SMA (functors + lattice, rarity-weighted). (c) Structure-mapping retrieval (MAC bound → FAC alignment), which yields a structural citation together with an abstain/novelty signal from expectation violation. (d) One universal loader mounts eleven ontologies spanning seven knowledge areas; a domain router routes across the evaluated domains without merging them.

a synthetic structure-only control, adds no measurable top-5 accuracy on the present real-domain tasks. We also delimit the thesis: where no discriminative subsumption ontology applies, the advantage disappears, and we report those nulls.

We make three contributions. First, a **universal ontology adapter** (loader + registry + router), demonstrated on eleven ontologies spanning seven knowledge areas with one shared retrieval core plus thin per-domain data adapters (gold and query loaders, no per-domain retrieval logic). Second, a **memory-swap benchmark** that holds the language model and prompt fixed and swaps only the retriever, evaluated against a strong RAG/KG baseline suite. Third, an **honest characterization** of when structure-mapping helps and when it does not.

2 Results

2.1 The universal adapter and the memory-swap protocol

The loader parses any OBO/OWL/STIX/CPC source into a normalized ontology graph, mounting the is-a tree as the ascension lattice and the typed relations as higher-order statements, and it indexes each entity's term-set as an analogical case (Table 1). Because a registry and a domain router select the relevant ontology per query, a single mechanism serves all five evaluated domains without ever merging them. The benchmark then holds the language model and prompt fixed and swaps only the retrieval memory among SMA and the baseline retrievers, so any difference is attributable to retrieval alone. The baseline suite comprises BM25 [5], BGE neural dense retrieval [4], hybrid reciprocal-rank fusion, hybrid with a cross-encoder reranker, and the graph retriever HippoRAG [6]. We score the rare/long-tail slice (entities whose rarest term has information content above the corpus median), where expertise is decisive, using a paired bootstrap (10^4 resamples) and Holm–Bonferroni correction across domains.

2.2 SMA matches a bespoke ontology oracle but outperforms general-purpose RAG/KG

A retriever that consumes the same ontology should be near-optimal on pure-subsumption ranking, and one is. Against Phenomizer [12], a hand-built ontology-aware information-content tool that scores the same Resnik [13] information content SMA uses, SMA reaches statistical parity on top-5 (HPO $\Delta=-0.033$, $p=0.037$, Holm $p=0.073$; GO $\Delta=-0.011$, $p=0.71$); it matches the bespoke oracle without beating it. Parity here does not license the stronger claim that higher-order structural alignment is what separates SMA from RAG; if anything, drawing even with an information-content oracle makes the opposite reading more plausible. The honest interpretation is that ontology-aware subsumption and information-content weighting do the heavy lifting (the mechanism ablation below confirms this directly): SMA beats the IC-agnostic BM25/dense/hybrid suite (Fig. 2, Table 2) because it mounts those signals as match geometry, and it merely ties the IC oracle that already has them. What makes SMA more than a domain-specific Phenomizer clone is therefore not superiority over information-content similarity but *generality*: the same retrieval core mounts multiple ontology formats and domains, whereas Phenomizer is a specialised biomedical oracle. We frame the oracle as a ceiling reference rather than an opponent, and state the headline claim as **SMA > RAG/KG**.

2.3 A structure-only control isolates the mechanism from lexical and information-content similarity

One concern is that SMA's advantage merely reflects information-content weighting or lexical overlap that a tuned retriever could recover. We rebut this directly with the Synthetic Structure Benchmark (SSB), an adversarial control in which structure is the *only* usable signal. Each query and its correct library entry are zero-lexical-overlap relational analogs bridged solely through the ascension lattice: they share no surface tokens, so BM25 and TF-IDF/dense retrieval are information-theoretically blind, and the gold match is recoverable only by aligning higher-order relational structure across the lattice. On the 200 pooled library queries (100 queries $\times$ 2 seeds), SMA attains rank-1 accuracy 0.895 (MRR 0.948), whereas both BM25 and TF-IDF/dense retrieval score 0.000 at rank-1 (MRR $\approx$ 0.010, at chance): $\Delta\, r@1$=+0.895, 95% CI $[0.850, 0.935]$, p_{Holm}=4×10^{-4}, Cliff's δ=0.895 against each lexical/dense baseline. In the forced-choice variant, where each query is paired against a single structural distractor, SMA reaches rank-1 1.000. The control is also de-circularized: we removed the earlier `far_`-prefix bijection and the $(i, i{+}2)$ rewiring, so no shortcut maps queries to entries by surface convention. Because SSB carries no information-content gradient for either retriever to exploit and is lexically orthogonal by construction, it serves as a *capability control*: it shows that, when relational alignment is the only usable signal, SMA's implemented higher-order structure-mapping solves the task while lexical and dense retrieval fail by construction. SSB is synthetic and adversarially constructed in SMA's favour, so we use it to establish that the machinery *can* exploit pure structure, not as evidence for the mechanism behind the real-domain benchmark gains; that attribution comes separately from the mechanism ablation below, which finds the real-domain advantage is carried by is-a subsumption and information content rather than higher-order relations (Extended Data Table 3).

2.4 Naive ancestor expansion does not explain the gain

A natural objection is that SMA wins merely because it sees each term's is-a ancestors, which the baseline retrievers do not. We tested this directly by giving BM25 and dense retrieval the same advantage: every document and query term-set was expanded with its full is-a ancestor closure before indexing (rare-slice top-5; Extended Data Table 4). Far from helping, ancestor expansion *hurts* the retrievers. On GO, BM25 falls from 0.776 to 0.504 and dense from 0.719 to 0.465; on HPO the same arms slip from 0.731 to 0.715 and from 0.622 to 0.561. The reason is that the shared common ancestors are high-frequency, low-information terms, and they dilute the IDF and cosine signal that the rare, decisive term would otherwise carry. SMA still beats every ancestor-expanded arm by Holm-significant margins (HPO +0.20 versus BM25+ancestors and +0.36 versus dense+ancestors; GO +0.34 and +0.38; all p_{Holm}=0.0012). The one honest nuance is on GO, where *unexpanded* BM25 reaches parity with SMA (Δ=+0.070, p_{Holm}=0.076, n.s.); on HPO, and on every expanded arm, SMA's lead is significant. This rules out the simplest text-side route to the same information, *naive* full-closure ancestor expansion, rather than every possible one: depth-limited or information-content-weighted expansion, parent-path encodings, separate ancestor fields, or learned sparse retrieval could narrow the gap and we have not tested them. What we can say is that bounded ascension over the mounted lattice captures the subsumption signal without the dilution that sinks naive expansion, and that the advantage is not an artifact of simply handing the ancestors to a bag-of-terms retriever.

2.5 A mechanism ablation isolates what structure-mapping adds

To attribute SMA's edge to its individual components, we added one mechanism at a time over a lexical-only base, on a rare-slice top-5 task (3 seeds, n_{index}=400, n_{query}=90, paired bootstrap of adjacent rungs; Extended Data Table 9). The advantage decomposes cleanly. On **medicine** (HPO, a pure-is-a ontology with no typed relations) is-a subsumption is the dominant lever: lexical-only 0.587 → +is-a 0.955 (Δ=+0.368, p=2×10^{-4}), after which information-content weighting adds a significant second increment (→ 0.982, Δ=+0.027, p=0.003). On **genomics** (GO, which does carry typed relations) the same ordering holds at a smaller scale: lexical 0.750 → +is-a 0.830 (Δ=+0.080, p=0.066) → +IC 0.910 (Δ=+0.080, p=2×10^{-4}). Critically, adding higher-order typed relations on top of is-a+IC yields *no* measurable top-5 gain (GO 0.910 → 0.915, Δ=+0.005, p=0.72, n.s.; HPO has no typed relations to add, Δ=+0.000). The advantage is thus carried by is-a subsumption (the dominant lever; HPO +0.368, GO +0.080) plus information content (a significant but smaller second increment), with typed higher-order relations adding nothing detectable at top-5. This is consistent with the Phenomizer IC parity and with the SSB control, where structural alignment is load-bearing precisely because no lexical or information-content signal is available. One caveat: the ablation runs at a reduced index scale (n_{index}=400, since full scale is compute-bound), so the robust quantity is the *relative* contribution of each rung, not the absolute accuracies (Extended Data Table 9).

2.6 Across the baseline retrieval suite, SMA wins on the rare tail

SMA leads on the rare/long-tail slice in every field we tested (Fig. 2, Table 2). On **medicine** (rare-disease diagnosis over the Human Phenotype Ontology [14] and MONDO [15]) SMA attains tail top-5 of 0.949 versus 0.606 for the best baseline configuration (hybrid + cross-encoder rerank), Δ=+0.333, 95% CI [0.281, 0.389], p<2×10^{-4}. On **genomics** (Gene Ontology [16, 17] gene-function assignment) SMA reaches 0.849 versus 0.682, Δ=+0.156, p<2×10^{-4}. On **finance** (SEC filing to US-GAAP concept) SMA attains 0.418 versus 0.231, Δ=+0.167, p<2×10^{-4}, roughly double the best RAG, though all methods score low here because US-GAAP concepts are weakly discriminative. On **legal** (patent to CPC classification, all-query slice) SMA reaches 0.941 versus 0.870, Δ=+0.064, p=0.002. These four domains are the load-bearing results: each survives Holm correction across fields and the conservative checks below. The same trend appears in **cyber** (MITRE ATT&CK [18] threat-group attribution), where SMA leads 0.766 versus 0.749 (Δ=+0.073), but this is the weakest arm and we do not treat it as a pillar: the margin is small, threat groups have distinctive technique vocabularies that lexical and dense retrieval partly capture, and as shown next the contrast does not survive conservative correction. Because "best RAG" is data-selected (chosen by top-5, then tested on top-5), we additionally compare SMA against *every* baseline in *every* domain. The advantage is strictly positive in all cases, and under a conservative Bonferroni-over-baselines correction four of five domains survive (medicine, genomics, finance at p_{Bonf}=0.0010, legal at 0.0110) while cyber is directional

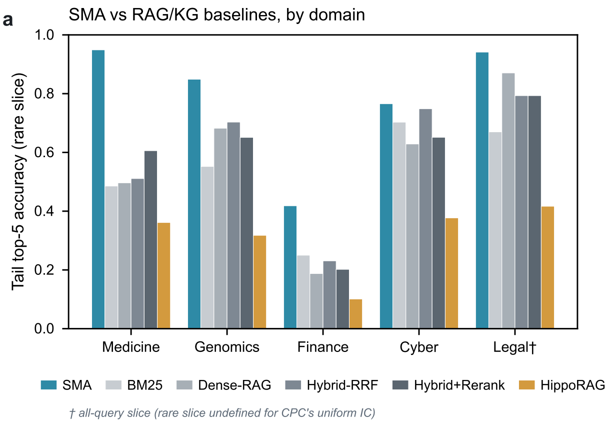

Fig. 2 **Agentic memory-swap results.** (a) Tail top-5 accuracy (rare slice; legal on the all-query slice) by domain: SMA versus the RAG/KG baseline suite. (b) Headline effect size Δ(SMA − best RAG) tail top-5 per domain with 95% CI; all above zero. AURC and structural-novelty F1 per domain are in Table 2(b) and Extended Data Fig. 1.

(p=0.035, p_{Bonf}=0.17; Extended Data Table 8). A second robustness check addresses the resampling unit. The headline bootstrap is per-query, but the three seeds share target entities, so we re-ran it as a *cluster bootstrap that resamples entities* (Methods; Extended Data Table 10). The four strong domains remain significant, whereas cyber weakens further: its delta is unchanged but the entity-clustered 95% CI now includes zero (p rises from 0.035 to 0.088). We therefore report cyber as a directional trend under both corrections. Legal's rare slice is undefined, because CPC's near-uniform information content yields no rare split, so legal is reported on the all-query slice and flagged as such.

2.7 Cite-or-abstain and structural novelty

SMA has the lowest risk-coverage area-under-the-risk-coverage-curve (AURC) in every one of the five arms (0.017–0.530 versus best-RAG 0.059–0.795). The reason is simple: a structural match either exists or it does not, whereas cosine similarity is always moderately high, so a pure-RAG retriever cannot tell when to abstain. On held-out unknown entities SMA's expectation-violation flag yields a structural-novelty F1 of $\approx$ 0.18 in every domain, matched only by the graph-based HippoRAG, while every pure-RAG baseline scores exactly 0. We report absolute novelty F1 honestly: the retrieval-suite threshold is fixed and untuned, so the contrast that matters is non-zero versus structurally zero rather than the absolute level (the end-to-end agent below calibrates the threshold per memory).

2.8 The end-to-end agent is more accurate, selective, and attributable than dense RAG

The retrieval suite isolates the retriever; we next place a real language model in the loop and measure what matters in deployment, where a confident wrong answer is catastrophic. Holding the language model (DeepSeek, temperature 0) and the prompt fixed and swapping only the memory (closed-book / dense-RAG / SMA), an ontology-grounded diagnosis agent answered 120 *answerable* (indexed-disease) and 120 *held-out* (unseen-disease) clinical cases, with a cite-or-abstain threshold calibrated per memory on a *disjoint* split (pre-registration v2; Methods). Across the trustworthy-QA composite the SMA-grounded agent dominates (Fig. 3). It is the most accurate (0.342 versus dense 0.100 versus closed-book 0.017; paired-bootstrap Δ=+0.242, 95% CI [0.167, 0.325], Holm-significant) and the most faithful in citation (ALCE-style support [19] 0.621 versus 0.480). Most decisive is that its raw structural grounding score separates known from unknown entities (AUROC 0.793 versus dense's near-chance 0.547; Δ=+0.246, [0.159, 0.333]). Because cosine similarity stays high even for unseen diseases, dense RAG can reach abstain-recall 0.90 only by refusing 79% of answerable cases, whereas SMA reaches 0.91 while refusing just 45%. Abstain-recall itself is therefore a *tie* (Δ=+0.008, p=0.92), and we report it as such. Net selective accuracy is 0.625 versus 0.500 (Δ=+0.125, [0.071, 0.179]), and structural-novelty F1 is 0.789 versus 0.553 (novelty-recall Δ=+0.308, [0.200, 0.408]); closed-book scores 0 on novelty and cannot cite at all. In sum, on this single-domain (medicine) case study four of five pre-registered axes are Holm-significant wins, abstain-recall is a tie, and the accuracy floor holds with SMA above rather than below RAG. At 0.342 top-1 exact-match accuracy the agent is a promising retrieval-grounded specialist rather than a finished one; but equipped with structure-mapping memory a generalist LLM answers more accurately, cites a checkable structural identifier, abstains discriminatively, and flags novelty, whereas equipped with vector RAG it answers less accurately and cannot tell when it does not know.

2.9 Negative results and boundary

We pre-registered a parity-reporting rule: where a task carries no discriminative subsumption ontology to mount, we report parity or a null rather than search for an edge. Two families satisfy this condition. The first is **flat single-record tabular prediction**. On hospital readmission and credit-card fraud, each record is encoded independently and the discriminative structure is cross-record, so SMA reaches parity with value-based retrieval and goes no further. We verified the mechanism directly: an LLM-drafted higher-order adapter mechanically raised relation density from 0, yet lifted SMA only to parity with dense retrieval, not past it. The second family arose from a follow-up boundary probe, in which we asked whether *graph* structure alone suffices when single-record structure does not. On the public Elliptic Bitcoin ledger [20] we encoded each transaction's graph neighbourhood (predecessor and successor flows) as higher-order relations over a licit/illicit typology lattice, with strict label-leak guards, supplying the cross-record structure that flat-tabular retrieval lacks and that graph neural networks [21] exploit. The outcome is an honest null. SMA shows no significant edge over dense or lexical retrieval (Δ accuracy +0.004, 95% CI [−0.010, +0.019], p=0.56), and every retrieval method, SMA included, trails a flat logistic-regression baseline on the full 166-feature record (macro-F1 0.787 versus SMA's 0.581; Extended Data Table 5). The lesson sharpens the thesis, because graph topology alone is not enough: a coarse licit/illicit typology yields too few distinct

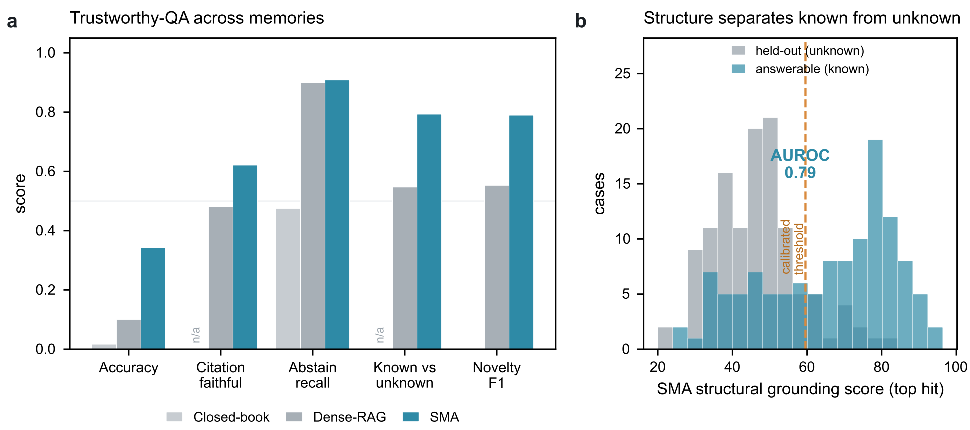

Fig. 3 The SMA-grounded LLM agent is more accurate, more selective, and more structurally attributable than dense RAG (medicine case study; n=120 answerable + 120 held-out; language model and prompt fixed, memory swapped). (a) Trustworthy-QA axes for closed-book / dense-RAG / SMA: accuracy, citation-faithfulness, abstain-recall, known-versus-unknown discrimination (grounding AUROC), and novelty F1; closed-book has no retrieval, so its citation and grounding cells are N/A (marked), not zero. (b) Mechanism: SMA's raw structural grounding score (top hit) separates answerable (known) from held-out (unknown) cases, AUROC 0.79; the dashed line is the cite-or-abstain threshold calibrated on a disjoint split. Dense cosine similarity barely separates the two (AUROC 0.55), forcing its gate to over-abstain.

structural signatures for analogical alignment to beat vector similarity, and structure helps SMA only when the subsumption ontology is itself richly differentiated, as in the curated (expert-maintained) ontologies. We draw the boundary rather than hide it. SMA needs a discriminative subsumption ontology, not mere graph topology.

3 Discussion

One mechanism serves medicine, genomics, finance, legal and cyber with a single shared retrieval core plus thin per-domain data adapters and no per-domain retrieval logic, a generality the bespoke ontology tools (one per domain) cannot match. The defensible asset is a registry of curated (expert-maintained) ontologies mounted as retrieval geometry; a language model can draft adapter rules but does not reproduce the expert curation encoded in the ontology. We are deliberately precise about which part of that geometry is load-bearing. Across the real ontology benchmarks, SMA's measured advantage is driven primarily by bounded is-a subsumption over the ontology lattice together with information-content weighting; the higher-order typed relations are implemented and are necessary for the synthetic structure-only control, but they do not measurably improve the current real-domain top-5 results (GO +0.005, n.s.; HPO and CPC carry no typed relations to add). Whether relation-rich ontologies would let higher-order structure-mapping pull ahead on tasks designed to require it is an open, untested prediction, not a result we claim here.

Several limitations bound these claims. The retrieval-suite novelty threshold is fixed and untuned, whereas the end-to-end agent calibrates it per memory on a disjoint split. HippoRAG is somewhat disadvantaged by a bag-of-term-names document representation that suits its phrase-graph poorly. The interactive end-to-end study is a single-domain (medicine) case study. Finally, we restrict to fully open ontologies and exclude licensed sources (SNOMED, UMLS) in favour of reproducibility.

3.1 What we do not claim

We state the boundaries of the thesis explicitly. **(i)** We claim no advantage on flat, single-record tabular prediction: hospital readmission and credit-card fraud reach only parity with value-based retrieval, and on the Elliptic Bitcoin ledger [20] SMA shows no significant edge over the best retrieval baseline (Δ accuracy +0.004, 95% CI $[-0.010, +0.019]$, p=0.56), with all retrieval methods trailing a flat logistic-regression baseline, since there is no ontology structure to exploit. **(ii)** We do not claim to beat the bespoke ontology oracle Phenomizer [12]: on pure-subsumption ranking SMA reaches parity only (HPO Δ=$-$0.033, GO Δ=$-$0.011; Holm n.s.), matching but not exceeding a hand-built information-content tool. The headline win is over ontology-agnostic RAG/KG, not over the oracle. **(iii)** The end-to-end LLM-in-the-loop study is medicine-only; the broader memory-swap results are retrieval-only. **(iv)** The retrieval-suite structural-novelty threshold is fixed and untuned, so its absolute novelty F1 is not a tuned operating point (only the end-to-end agent calibrates the threshold per memory). **(v)** We restrict to fully open ontologies and exclude licensed vocabularies (SNOMED, UMLS) in favour of reproducibility, so we make no claim about coverage those sources would add. **(vi)** This is a *representational* claim, namely structure as retrieval geometry, and not a claim about model scale or fine-tuning; the language model and prompt are held fixed throughout and only the memory is swapped. **(vii)** In our mechanism ablation, higher-order typed relations did not add measurable tail top-5 accuracy beyond is-a subsumption combined with information content; the gain we report is attributable to subsumption and IC weighting, not to relations over relations (Extended Data Table 9).

3.2 Relation to prior work

Grounding LLM memory in ontologies is an active 2024–2026 area; our contribution is not ontology-grounding per se but the structure-mapping mechanism and the universal adapter that mounts a broad class of curated ontologies as match geometry. Information-content semantic-similarity measures (Resnik [13]; Lin [22]; Jiang–Conrath [23]) score term-set overlap weighted by corpus frequency, whereas SMA aligns relational statements $r(f_s(x), f_o(x))$ over the ascension lattice. The IC oracle Phenomizer [12] uses this same information content; that SMA ties it (HPO Δ=$-$0.033, GO Δ=$-$0.011; parity after Holm) while beating the IC-agnostic retrieval suite indicates that ontology-aware subsumption and information content, mounted as match geometry, are what separate SMA from general-purpose RAG, rather than superiority over information-content similarity itself. The ontology- and knowledge-graph-aware retrievers, namely OG-RAG [24] and OntologyRAG [25] together with the KG-RAG line of GraphRAG [7] and HippoRAG-2 [26], flatten the ontology into retrievable text or an OpenIE-derived phrase graph; SMA instead mounts the asserted lattice directly as typed match geometry, with no OpenIE step and no per-domain retrieval code. Finally, unlike case-based reasoning over LLM memories (CBR-RAG [27]), where retrieval reduces to surface feature similarity, SMA's first stage is structural alignment under an admissible MAC bound [10]. Specialised biomedical tooling (the Ontology Access Kit [28]; Exomiser [29]) realises ontology-aware similarity within single domains, whereas SMA supplies one shared mechanism across all of them.

4 Methods

4.1 Representation and matching

Statements are functors over entities; SME builds consistent kernel mappings [9]; MAC/FAC provides a certified admissible content bound for tractable first-stage retrieval and best-first alignment for the second [10]. Cross-vocabulary matching ascends the ontology lattice with penalty ρ^{dist} (frozen ρ=0.95, ascension depth δ=2). The surprisal scorer weights a matched functor by its information content $-\log_2 p$. End to end, the universal adapter and matcher pipeline (Fig. 4) ingests a broad class of open ontologies into a normalized graph, mounts each as match geometry, and registers the resulting cases behind a per-query domain router.

4.2 Cases as functored statements

The loader encodes each subject (an entity record) as a *case*, a set of logical statements. Each present ontology term T on a subject x is the unary statement $f_T(x)$, and each asserted typed relation (s, r, o) whose endpoints s and o are both present on x becomes the higher-order statement $r(f_s(x), f_o(x))$, a relation over relations. A case is the set of all such statements for x. Here "functor" denotes a predicate / term-constructor symbol in the structure-mapping sense (it labels a statement and may take statements as arguments) and is *not* a category-theory functor; this higher-order constructor is what lets SME align relations between relations [9]. Subsumption is not materialized at load time: the loader extracts the asserted is-a DAG plus the typed relations into a single normalized graph, and generalization is realized only at match time by bounded δ-hop ascension over asserted edges (no OWL reasoner is invoked). Multi-parent DAGs are supported, cyclic is-a chains are truncated at depth δ, and OWL axioms beyond `subClassOf` and existential / universal (`someValuesFrom` / `allValuesFrom`) restrictions are not consumed.

4.3 Ascension and information content

A specific term matches a more general one along the lattice with factor

$$a(m) = \rho^{\,\mathrm{dist}(m)}, \qquad \mathrm{dist}(m) \le \delta, \tag{1}$$

where $\mathrm{dist}(m)$ is the minimal is-a distance between the matched terms and a match beyond δ hops is disallowed (frozen ρ=0.95, δ=2). Information content is the Resnik information-content estimate [13] from corpus term frequency with is-a closure propagation,

$$\mathrm{IC}(c) = -\log_2 \frac{\mathrm{count}(c)}{n}, \tag{2}$$

where $\mathrm{count}(c)$ accumulates every annotation of c and of its is-a descendants and n is the corpus total; the surprisal scorer sets a matched functor's base weight to its surprisal $\sigma_0(m) = -\log_2 p(m)$ (and to 1 when the functor is uncosted), so the rare-but-decisive term dominates by construction.

4.4 Trickle-down structural score

The structural support of a case is the sum over match hypotheses m of a trickle-down score in which a hypothesis inherits a discounted share of the support of the hypotheses it is an argument of,

$$S(m) = \sigma_0(m)\, a(m) + \gamma \sum_{p \in \mathrm{parents}(m)} S(p), \tag{3}$$

with discount γ=0.25; $\mathrm{parents}(m)$ are the hypotheses whose aligned arguments are m, so systematic (deeply nested) structure compounds while isolated term matches do not. Setting the functor costs to none recovers a uniform-weight structural evaluation ($\sigma_0 \equiv 1$).

4.5 Admissible content bound

First-stage MAC retrieval ranks candidates by an admissible upper bound on the structural score that never underestimates the true alignment, so no true match is pruned:

$$U(q, c) = \bar{s} \sum_f \mathrm{cost}(f)\, \min\bigl(q_f, c_f\bigr), \tag{4}$$

where q_f and c_f are the functor-f counts of the query and candidate content vectors and $\bar{s}$=4.0 is the per-match-hypothesis score ceiling in production. Only the bounded shortlist is passed to the FAC alignment stage [10].

4.6 Novelty by expectation violation

A held-out case is flagged as novel by its expectation violation against the indexed schemas,

$$\mathrm{ev}(\mathrm{case}) = 1 - \max_g \, \mathrm{fit}_{\mathrm{norm}}\bigl(\mathrm{case}, \mathrm{schema}_g\bigr), \tag{5}$$

where $\mathrm{fit}_{\mathrm{norm}} \in [0, 1]$ is the normalized structural-match score of the case against learned generalization g; ev = 1 when no generalization yet exists (nothing is expected). Novelty is thus detected from structural mismatch, not from term absence.

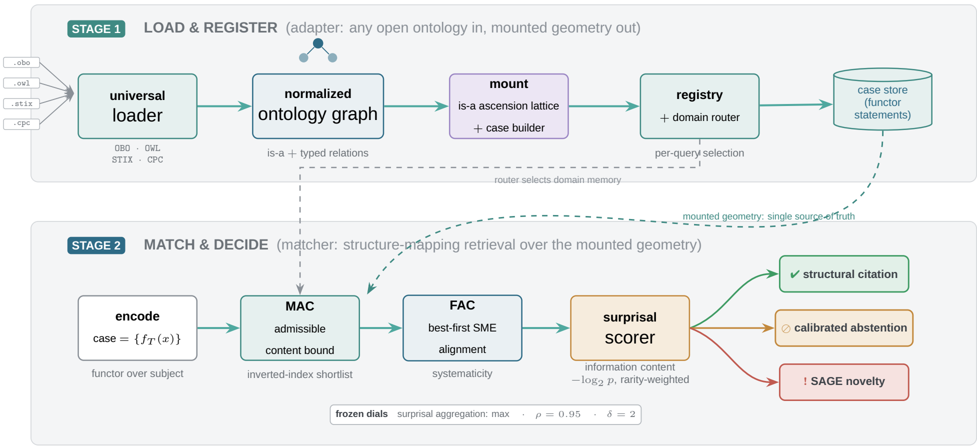

Fig. 4 SMA architecture. A universal loader ingests any open ontology (OBO/OWL/STIX/CPC) into a normalized graph, mounts the is-a hierarchy and typed relations as match geometry, builds higher-order cases, and registers them behind a per-query domain router. At query time a case (functors over a subject) is screened by a certified admissible-content (MAC) bound, aligned by best-first structure-mapping (FAC/SME), scored by rarity-weighted information content, and resolved to a structural citation, a calibrated abstention, or a novelty flag.

4.7 Citation faithfulness

Citation faithfulness is computed automatically as an exact structural identifier match: among answerable items that the agent answered (did not abstain on) and cited, the fraction whose cited candidate identifier equals the gold identifier. It is therefore an ALCE-style support [19] *proxy* based on structural attribution, not an entailment / NLI judgement. Because it conditions on answered items only, it is reported alongside but not on the same denominator as accuracy (e.g. 0.621 faithfulness over cited answers versus 0.342 accuracy over all answerable cases).

4.8 Language-model decoding and failure semantics

The end-to-end agent queries DeepSeek (`deepseek-chat`) with temperature 0, `max_tokens`=600, a single attempt, a 60 s timeout and no retries; the temperature-0.3 verbalizer path is not exercised by the QA harness. Each reply is parsed as strict one-line JSON, and any parse, type or range failure collapses deterministically to ABSTAIN, the safe action, so a malformed completion is never scored as an answer.

4.9 Universal adapter

The loader (`sma/ontology`) provides `load_obo`, `load_owl`, `load_owl_dir`, `load_rdflib`, `load_attack_stix`, `load_cpc` and `load_mitre_xml`; `mount()` builds the lattice and the case builder; a registry and a domain router select the ontology per query. Ontologies are version-pinned, and `PYTHONHASHSEED=0` with sorted set→list conversion ensures process-independent reproducibility.

4.10 Memory-swap retrieval benchmark

The harness (`sma/eval/agentic`) wraps SMA and each baseline behind one interface, indexes identical records, generates hard partial/imprecise/noisy queries, and scores tail top-k (rare slice = rarest-term information content above the corpus median), risk-coverage AURC, and structural-novelty F1, with a paired bootstrap (10^4 resamples, seed 12345) and Holm–Bonferroni correction. Baselines: BM25 [5]; BGE-small neural dense [4]; hybrid reciprocal-rank fusion; hybrid + `bge-reranker-base`; and HippoRAG-2 [6].

4.11 Entity-clustered robustness and computational cost

The headline paired bootstrap resamples per query, but each target entity contributes up to three correlated queries (one per seed), so per-query resampling is anti-conservative at the entity level. As a robustness check we re-ran the identical evaluation with a *cluster bootstrap* that resamples target entities with replacement, taking all of a resampled entity's queries together (`scripts/cluster_bootstrap_suite.py`; the per-query point estimate is reproduced exactly, then re-aggregated by entity). We report the entity-clustered Δ, 95% CI and p alongside the per-query values in Extended Data Table 10. Separately, we measured computational cost on the same hardware (`scripts/cost_latency.py`): index-build time, per-query latency (p_{50}/p_{95}) and peak resident memory for SMA, BM25 and dense retrieval, with each arm timed in its own process (Extended Data Table 11). SMA uses no learned embeddings, no GPU and no vector store; the trade-off it pays is a slower, CPU-bound build and a higher per-query latency than lexical BM25.

4.12 End-to-end LLM-QA agent

The agent (`sma/eval/agentic_qa`, pre-registration v2) holds DeepSeek (temperature 0) and the prompt fixed, swaps the memory (closed-book / dense-RAG / SMA), retrieves top-k, then answers and cites or abstains. The cite-or-abstain signal is the raw top grounding score, gated at a threshold calibrated per memory on a disjoint split (Youden's J, retrieval-only, no LLM spend), below which the agent abstains and flags novelty. Accuracy is scored against the gold diagnosis; citation faithfulness uses an ALCE-style support check [19]; abstention is summarized by risk-coverage and selective accuracy; known-versus-unknown discrimination is the threshold-free grounding AUROC. Here *verifiable* means that every answer carries a checkable structural citation (an exact structural match of the cited candidate to the gold case) together with a calibrated abstention signal. Because the SMA grounding score (summed information content, in bits; threshold ≈ 59.6) and the dense cosine score ($[0, 1]$; threshold ≈ 0.82) live on incomparable scales, raw scores cannot be thresholded across memories; we therefore (i) calibrate each gate per memory on the disjoint split and (ii) report the scale-free grounding AUROC as the headline known-vs-unknown metric. Per-axis paired bootstraps and Holm correction follow the pre-registration.

Data and code availability. *Code.* The universal ontology adapter, the memory-swap retrieval harness, the end-to-end LLM-QA agent, and all analysis and figure

Table 1 The eleven mounted curated (expert-maintained) ontologies (one universal loader).

Domain	Ontology	Terms	is-a	Typed rel.
Medicine	HPO	19,810	24,329	0
Medicine	MONDO	56,273	78,736	34,157
Anatomy	Uberon	14,973	19,382	11,697
Biology	GO	38,263	57,803	14,076
Chemistry	ChEBI	205,592	285,589	94,902
Cyber	MITRE ATT&CK	712	475	872
Cyber	MITRE CAPEC	558	532	194
Cyber	MITRE CWE	944	1,160	284
Legal/IP	CPC	254,274	256,106	0
Legal	LKIF-core	153	171	48
Finance	FIBO	2,948	3,819	1,335
Total		**≈594k**		

Table 2 Agentic memory-swap results across five domains. (a) Tail top-5 (rare slice; legal on the all-query slice †, since CPC's near-uniform information content yields no rare split). (b) Capability axes (AURC lower is better; novelty F1). Bold = best per column. Δ and p are the per-query paired-bootstrap mean and Holm-adjusted significance versus the committed best baseline, so Δ need not equal the difference of the marginal accuracies shown (most visibly for cyber).

(a) Tail top-5 accuracy (rare slice) Memory	Medicine	Genomics	Finance	Cyber	Legal†
SMA (ours)	**0.949**	**0.849**	**0.418**	**0.766**	**0.941**
BM25	0.485	0.552	0.250	0.703	0.670
BGE dense	0.496	0.682	0.188	0.629	0.870
Hybrid-RRF	0.511	0.703	0.231	0.749	0.793
Hybrid + rerank	0.606	0.651	0.202	0.651	0.793
HippoRAG (KG)	0.361	0.318	0.101	0.377	0.417
Δ SMA − best RAG	+0.333	+0.156	+0.167	+0.073	+0.064
(p, Holm)	$< 6\times10^{-4}$	$< 4\times10^{-4}$	$< 2\times10^{-4}$	0.035	0.002
(b) Risk-coverage AURC / novelty F1					
SMA	**0.017**/0.18	**0.106**/0.18	**0.530**/0.18	**0.096**/0.18	**0.021**/0.18
best RAG	0.317/0.00	0.298/0.00	0.795/0.00	0.261/0.00	0.059/0.00
HippoRAG	0.628/0.19	0.721/0.19	0.925/0.18	0.560/0.18	0.462/0.18

scripts are released under the Apache-2.0 licence at https://github.com/ayazkhan27/SMA-1. The version supporting this paper is the `v1.0.0` release tag of that repository; the exact commit is recorded at the tag and snapshotted at Zenodo through the repository's GitHub integration. Frozen sub-artifacts are additionally tagged `adapter-v1`, `ontology-v1`, `score-v2` and `prereg-v1`; the end-to-end LLM-QA study is registered in `configs/preregistration_v2_llmqa.md` and all pre-registrations are committed before the corresponding runs.

Data and licensing. All mounted ontologies are openly licensed and version-pinned; their release versions and file `md5` checksums are recorded in `data/manifests/datasets.json` (e.g. GO `releases/2026-05-19`, MONDO `releases/2026-06-02`, Uberon `releases/2026-04-01`). Per-source licences are given in Extended Data Table 7 (HPO, GO, MONDO, Uberon, ChEBI, Orphanet and the GO Annotation File under CC BY 4.0; MITRE ATT&CK under Apache-2.0; CPC and US-GAAP in the public domain). Benchmark gold labels derive from public sources (the HPO disease–phenotype annotation file, the GO Annotation File, SEC EDGAR filings, USPTO patent CPC assignments, and the STIX ATT&CK corpus). We redistribute only derived gold labels (disease identifiers and phenotype/annotation sets), not OMIM source records (Extended Data Table 7). Confirmatory result tables, per-query bootstrap outputs, and the calibration grid are committed as comma-separated files under `reports/confirmatory/`.

Reproducibility. Runs are deterministic: `PYTHONHASHSEED=0` with sorted set→list conversion, the paired bootstrap seeded at 12345, and frozen ontology snapshots. The reference environment recorded with the release is Python 3.10.12 with NumPy 2.0.2, scikit-learn 1.5.2, sentence-transformers 5.2.2 and Google OR-Tools (full pins in `pyproject.toml` and the lockfile). All retrieval-only experiments, including the cost/latency benchmark, ran CPU-only on an AMD Ryzen 7 6800H (16 threads, 14 GiB RAM, Linux); no GPU is required. The end-to-end agent additionally calls DeepSeek (`deepseek-chat`, temperature 0); retrieval-only results require no LLM access.

Figure reproducibility. The conceptual figures Fig. 1 (overview) and Fig. 4 (architecture) are vector TikZ sources under `paper/figures/tikz/`; the data figures Fig. 2 (memory-swap results) and Fig. 3 (trustworthy-QA) are regenerated from the confirmatory CSVs by `scripts/figures_paper.py` and Extended Data Figs. 1–3 by `scripts/figures_ed.py`. The cost/latency and entity-clustered statistics are produced by `scripts/cost_latency.py` and `scripts/cluster_bootstrap_suite.py`.

Archival DOI. A snapshot of the `v1.0.0` commit is archived at Zenodo through the repository's GitHub integration; the DOI will be added on acceptance, and no DOI is asserted at submission time.

Extended per-domain metrics, the cross-system transfer study, and risk-coverage curves appear in the Supplementary Information.

Competing interests. The author declares no competing interests.

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Extended Data

Leakage regime: out-of-index, in-ontology

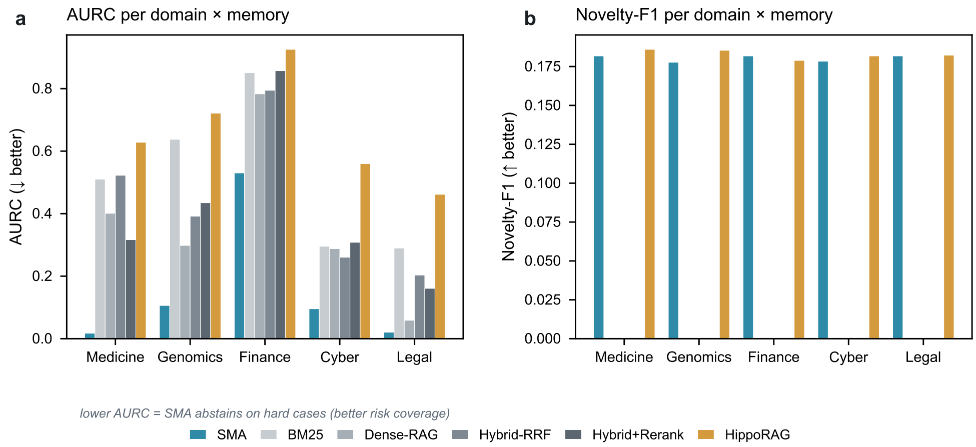
Held-out cases are *out-of-index* (the entity's case record is removed from the indexed pool) but *in-ontology* (its term classes remain in the mounted lattice). Unknown and known cases therefore draw from the same vocabulary, the harder and more realistic regime, rather than the trivial setting in which novelty coincides with unseen terms. Novelty is detected from *expectation violation* against the indexed cases (the structural grounding score has no analogue to align to), not from term absence. The rare-by-information-content threshold (rarest-term IC above the corpus median) is computed over the *indexed pool only*, so the slice definition never consults held-out cases and no test-on-train leakage can inflate the rare-slice contrast.

Statistical rigor: multiplicity, resampling units, and selection

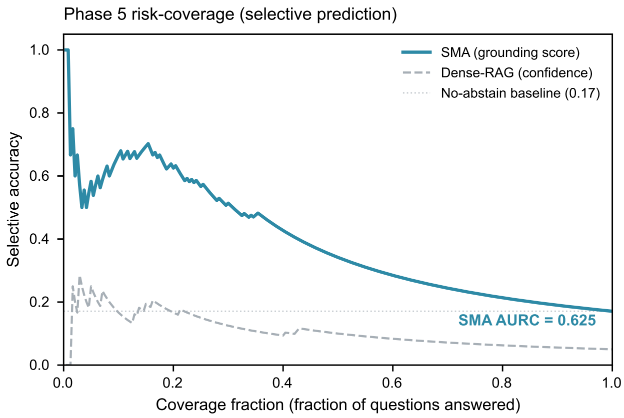
Two Holm families. We apply Holm–Bonferroni correction within two pre-registered families: the *retrieval-suite* family (across the five evaluated domains: medicine, genomics, finance, cyber, legal) and the *end-to-end* family (across the five trustworthy-QA axes: accuracy, abstain-recall, novelty-recall, selective-accuracy, grounding-AUROC).

Resampling unit. The retrieval-suite paired bootstrap (10^4 resamples, seed 12345) is *per-query, not per-entity*: the three seeds share the same underlying entities, so the effective number of independent entities is roughly one-third of the resampled query count. We disclose this because per-query resampling treats correlated queries over a shared entity as independent and therefore yields anti-conservative confidence intervals; the end-to-end design avoids this (its 120 answerable and 120 held-out cases are distinct diseases, one query each, and the bootstrap resamples entities, with the grounding-AUROC bootstrap resampling the positive and negative pools separately).

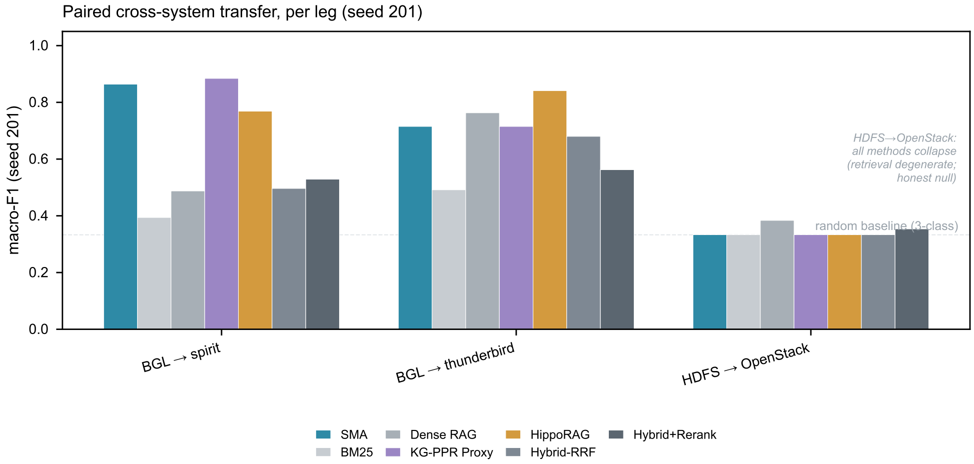
Selection correction (D5). The headline comparator "best RAG" is *data-selected*: the strongest baseline is chosen by all-query top-5 and then tested on the same statistic, which double-dips. To address this directly we compare SMA against *every* one of the five baselines in *every* domain (`agentic_vs_all_baselines.csv`). SMA's tail top-5 advantage is strictly positive against all five baselines in all five domains, so the "best-RAG" win is not a selection artifact. Applying a conservative Bonferroni-over-baselines correction ($\times 5$) to the vs-best p-value, *four of five domains survive*: medicine, genomics and finance at $p_{\mathrm{Bonf}} = 0.0010$ and legal at $p_{\mathrm{Bonf}} = 0.0110$. Cyber does not survive the selection correction ($p = 0.035 \rightarrow p_{\mathrm{Bonf}} = 0.17$) and we report it as *directional*, consistent with threat groups' distinctive technique vocabularies, which lexical and dense retrieval partly capture.



Extended Data Fig. 1 | Per-domain AURC and Novelty-F1 across all six memory systems. a, AURC (area under the risk-coverage curve; lower is better): SMA achieves the lowest AURC in all 5 of 5 domains, indicating superior selective abstention — it correctly withholds on hard or novel queries. Finance is the hardest domain for all methods (sparse ontology coverage). **b**, Novelty-F1: only SMA and HippoRAG return non-zero novelty-F1; the four lexical/dense baselines cannot distinguish known from unknown at all (novelty-F1 = 0). SMA novelty-F1 ranges from 0.177 (Genomics) to 0.182 (Medicine, Finance, Legal), reflecting the fixed 36-novel-query injection across arms. Source: `reports/confirmatory/agentic_{domain}.csv`.



Extended Data Fig. 2 | Phase 5 selective-prediction (risk-coverage) curve. Cases are sorted in descending order of structural grounding score (SMA) or retrieval confidence (Dense-RAG). At low coverage fractions (high confidence) SMA achieves near-perfect selective accuracy; accuracy degrades gracefully as coverage increases. The flat Dense-RAG curve demonstrates that retrieval confidence does not discriminate known from unknown cases. AURC (SMA) = 0.6253 vs Dense-RAG = 0.8248 (lower = better). Source: `reports/confirmatory/qa_sma.csv`, `qa_dense.csv`, `qa_sma_summary.csv`.



Extended Data Fig. 3 | Paired cross-system transfer, per leg (seed 201). macro-F1 for each transfer leg of the LogHub benchmark (`t1_transfer_metrics.csv`), comparing SMA against all baselines without leg-averaging (which masks per-leg imbalance). BGL → Thunderbird: SMA achieves 0.715 macro-F1. BGL → Spirit: SMA is competitive but one seed shows large variance (seeds 201–205 are shown in aggregate stats in `t1_stats.csv`). HDFS → OpenStack: all methods collapse to 0.333 (retrieval degeneracy; three-class uniform prediction; documented diagnostic in the CSV). The honest null on HDFS → OpenStack is reported and does not affect the main claims. Source: `reports/confirmatory/t1_transfer_metrics.csv`.

Extended Data Table 1 | Full per-domain metrics across all six memory systems. Tail top-5 accuracy (all-query and rare-query slices), area under the risk-coverage curve (AURC; *lower = better selective abstention*), and novelty-F1 for each memory × domain combination. Legal has no rare slice (uniform information content in CPC hierarchy; denoted *n/a*). All numbers from `reports/confirmatory/agentic_{domain}.csv`.

Domain	**Memory**	**Top-5 (all)**	**Top-5 (rare)**	**AURC↓**	**Novelty-F1**
Medicine	SMA	0.9537	0.9489	0.0174	0.1818
	BM25	0.4630	0.4854	0.5104	0.0000
	Dense-RAG	0.4815	0.4964	0.4010	0.0000
	Hybrid-RRF	0.5062	0.5109	0.5226	0.0000
	Hybrid+Rerank	0.5833	0.6058	0.3166	0.0000
	HippoRAG	0.3580	0.3613	0.6282	0.1860
Genomics	SMA	0.7963	0.8490	0.1061	0.1777
	BM25	0.4352	0.5521	0.6376	0.0000
	Dense-RAG	0.6235	0.6823	0.2982	0.0000
	Hybrid-RRF	0.6204	0.7031	0.3918	0.0000
	Hybrid+Rerank	0.5401	0.6510	0.4348	0.0000
	HippoRAG	0.2778	0.3177	0.7214	0.1854
Finance	SMA	0.3765	0.4183	0.5302	0.1818
	BM25	0.1821	0.2500	0.8508	0.0000
	Dense-RAG	0.1698	0.1875	0.7832	0.0000
	Hybrid-RRF	0.1914	0.2308	0.7945	0.0000
	Hybrid+Rerank	0.1728	0.2019	0.8572	0.0000
	HippoRAG	0.0741	0.1010	0.9254	0.1789
Cyber	SMA	0.8108	0.7657	0.0957	0.1784
	BM25	0.6441	0.7029	0.2956	0.0000
	Dense-RAG	0.6306	0.6286	0.2879	0.0000
	Hybrid-RRF	0.7297	0.7486	0.2605	0.0000
	Hybrid+Rerank	0.6622	0.6514	0.3081	0.0000
	HippoRAG	0.4189	0.3771	0.5600	0.1818
Legal†	SMA	0.9414	*n/a*	0.0206	0.1818
	BM25	0.6698	*n/a*	0.2899	0.0000
	Dense-RAG	0.8704	*n/a*	0.0587	0.0000
	Hybrid-RRF	0.7932	*n/a*	0.2035	0.0000
	Hybrid+Rerank	0.7932	*n/a*	0.1608	0.0000
	HippoRAG	0.4167	*n/a*	0.4617	0.1823

†Legal: rare slice undefined (CPC hierarchy has near-uniform information content; $n_{\text{rare}} = 0$ out of 324 queries). Top-5 (all-query) used as primary metric. **Bold** entries indicate SMA holds the best AURC in 5/5 domains and highest top-5 (all) in 5/5 domains.

Extended Data Table 2 | Phase 5 trustworthy-QA metrics and paired-bootstrap deltas. Three memory conditions: Closed-book (no retrieval), Dense-RAG, and SMA. The five trustworthy-QA axes are: accuracy (exact-match), citation faithfulness, abstain recall (correctly withholding on held-out unknowns), grounding-AUROC (structural score separates known from unknown), and novelty-F1 (detects out-of-memory queries). The *SMA vs Dense* column shows Δ = SMA − Dense with 95% bootstrap CI and Holm-adjusted p-values from `qa_stats.csv`. n/a = metric not applicable (closed-book has no retrieval scores).

Metric	**Closed-book**	**Dense-RAG**	**SMA**	**Δ (SMA−Dense)**	**Holm p**
Accuracy	0.017	0.100	0.342	$+0.242$ $[+0.167, +0.325]$	$< 0.001^{***}$
Citation faithfulness	n/a	0.480	0.621	—	—
Abstain recall	0.475	0.900	0.908	$+0.008$ $[-0.058, +0.075]$	0.917 (n.s.)
Grounding AUROC	n/a	0.547	0.793	$+0.246$ $[+0.159, +0.333]$	$< 0.001^{***}$
Novelty-F1	0.000	0.553	0.789	$+0.308$ $[+0.200, +0.408]$	$< 0.001^{***}$
Selective accuracy	0.246	0.500	0.625	$+0.125$ $[+0.071, +0.179]$	$< 0.001^{***}$
Score threshold	n/a	0.821	59.58	—	—
AURC	0.973	0.825	0.625	—	—

Closed-book values from `qa_none_summary.csv`; Dense-RAG from `qa_dense_summary.csv`; SMA from `qa_sma_summary.csv`; Δ and CI from `qa_stats.csv` (5 000 paired-bootstrap resamples). ***Holm-adjusted $p < 0.001$; n.s.=not significant. Citation faithfulness not applicable to closed-book (no retrieved passages). Abstain recall: 4/5 axes Holm-significant; abstain-recall is the honest tie (both systems achieve ≈90% recall of held-out unknowns).

Extended Data Table 3 | Synthetic Structure Benchmark (SSB): a structure-only control. Zero-lexical-overlap, lattice-bridged relational analogs where the gold match is recoverable only by structure-mapping; BM25 and TF-IDF/dense retrieval are information-theoretically blind by construction. *Library* task: n=200 pooled queries (100 queries × 2 seeds, seeds 41/43); *forced-choice*: each query against a single structural distractor. Deltas, 95% CIs, Holm-adjusted p-values, and Cliff's δ are SMA versus each baseline (paired bootstrap). The control is de-circularized (no `far_`-prefix bijection; no $(i, i{+}2)$ rewiring). All numbers from `reports/confirmatory/ssb_stats.csv` and `ssb_summary.csv`.

Task	**Method**	**r@1**	**MRR**	**$\Delta\, r$@1**	**95% CI**	**p_{Holm}**	**Cliff's δ**
Library	**SMA (ours)**	**0.895**	**0.948**	—	—	—	—
	BM25	0.000	0.010	+0.895	$[0.850, 0.935]$	$4{\times}10^{-4}$	0.895
	TF-IDF/dense	0.000	0.010	+0.895	$[0.850, 0.935]$	$4{\times}10^{-4}$	0.895
Forced-choice	**SMA (ours)**	**1.000**	**1.000**	—	—	—	—

r@1 = rank-1 accuracy; MRR = mean reciprocal rank. BM25 and TF-IDF/dense score 0.000 at rank-1 (MRR ≈ 0.010, at chance) because query and gold entry share no surface tokens; the match is recoverable only by aligning higher-order relational structure across the ascension lattice. SSB carries no information-content gradient for either baseline to exploit, so the result isolates SMA's structure-mapping mechanism from both lexical overlap and information-content similarity.

Extended Data Table 4 | Ancestor-expanded RAG and information-content baselines (rare-slice top-5). Each baseline indexes the byte-identical flattened term-name record; the +ancestors arms additionally expand every document and query term-set with its full is-a ancestor closure before indexing. Phenomizer (IC best-match average) and Jaccard are ontology-aware information-content references. Δ = SMA − method on the rare-query slice; p_{Holm} is Holm–Bonferroni-corrected across baselines within each ontology (10^4 paired-bootstrap resamples). Giving BM25/dense the is-a ancestor closure *lowers* their accuracy (common ancestors dilute IDF/cosine); SMA beats every expanded RAG arm. The sole non-significant contrast is plain BM25 on GO. All numbers from `reports/confirmatory/ontology_fair_baselines.csv`.

	HPO (rare disease)			GO (gene function)		
Method	**Top-5**	Δ	p_{Holm}	**Top-5**	Δ	p_{Holm}
SMA (ours)	**0.917**	—	—	**0.846**	—	—
Phenomizer (IC-BMA)	0.952	−0.035	0.047	0.860	−0.013	0.677
Jaccard	0.708	+0.208	0.0012	0.737	+0.110	0.0012
BM25	0.731	+0.186	0.0012	0.776	+0.070	0.076
BM25 + ancestors	0.715	+0.202	0.0012	0.504	+0.342	0.0012
Dense-RAG (TF-IDF)	0.622	+0.295	0.0012	0.719	+0.127	0.0012
Dense-RAG + ancestors	0.561	+0.356	0.0012	0.465	+0.382	0.0012

Phenomizer is an information-content oracle reference (negative Δ: SMA reaches parity but does not beat it; see main text and Extended Data Table 1). The +ancestors arms are the fairness control: ancestor-expanding the retrievable text *reduces* BM25 and dense accuracy on both ontologies, so SMA's margin is not an artifact of ancestor visibility. The only non-significant SMA contrast is against unexpanded BM25 on GO (p_{Holm}=0.076).

Extended Data Table 5 | Elliptic Bitcoin graph-fraud boundary probe (honest null). SMA, dense, and lexical retrieval over a graph-neighbourhood licit/illicit typology encoding, versus a flat logistic-regression baseline on the full 166-feature record. SMA shows no edge over the best retrieval baseline (Δ accuracy +0.004, 95% CI $[-0.010, +0.019]$, p=0.56), and *all* retrieval methods trail logistic regression. Macro-F1 averages over the licit and illicit classes; illicit-F1 isolates the rare fraud class (219 illicit of 2,250 pooled test instances); ROC-AUC is threshold-free. All numbers from `reports/confirmatory/agentic_fraud_elliptic.csv` (k=15; seeds 7|17|23).

Method	**Macro-F1**	**Illicit-F1**	**ROC-AUC**
SMA (graph typology)	0.581	0.243	0.622
Dense-RAG	0.586	0.262	0.727
BM25	0.602	0.289	0.709
Logistic regression (flat, 166-feat.)	**0.787**	**0.621**	**0.919**

Bold = best per column. The flat logistic-regression baseline on the native 166 engineered transaction features outperforms every structure/retrieval method, including SMA, on all three metrics: on this task the discriminative signal is in the per-record feature vector, not in a coarse typology lattice over the transaction graph. SMA vs. best retrieval baseline: Δ accuracy +0.004, 95% CI $[-0.010, +0.019]$, p=0.56 (per-query paired bootstrap; the three seeds share entities, so the effective N is $\approx 3\times$ smaller than the 2,250 pooled rows). Source: `reports/confirmatory/agentic_fraud_elliptic.csv`.

Extended Data Table 6 | Datasheet for the memory-swap benchmark (Gebru-style). Per-arm data source, licence, eligibility band, and pooled query counts; the shared query generator, indexing, and fairness protocol are stated in the notes. Sources are version-pinned; all counts are from `reports/confirmatory/agentic_{domain}.csv` and the arm loaders in `sma/eval/agentic/arms/`.

Arm	**Entity → gold**	**Source**	**Licence**	**Band**	$n_{\mathbf{all}}$	$n_{\mathbf{rare}}$	$n_{\mathbf{novel}}$
Medicine	disease → HPO phenotypes	`phenotype.hpoa` (HPO/OMIM/Orphanet)	CC BY 4.0‡	7–30	324	274	36
Genomics	protein → GO processes	`goa_human.gaf` (GOA)	CC BY 4.0	7–30	324	192	36
Finance	filing → US-GAAP concepts	SEC FSDS `num.txt` (2024Q1)	Public domain	10–40	324	208	36
Cyber	threat group → ATT&CK TTPs	ATT&CK STIX `enterprise-attack.json`	Apache-2.0	7–30	222	175	24
Legal/IP	patent → CPC subgroups	USPTO grant XML `ipg161011.xml`	Public domain	7–30	324	*n/a*†	36

Eligibility. An entity is eligible only if its gold term-set falls in the arm's band (`MIN_TERMS–MAX_TERMS`): 7–30 for medicine/genomics/cyber/legal, 10–40 for finance. Below the floor a case is trivial; above the ceiling SME kernel enumeration becomes intractable (excluding e.g. the most-documented threat groups). Bands are applied identically to all memories.

Query generator (`harness.py:make_qspec`, shared by every arm and every memory). From an entity's eligible term-set: (i) sample at most 5 gold terms (`rng.sample(terms, min(5, len(terms)))`); (ii) for each kept term, climb 0–2 is-a levels by `rng.choice([0,0,1,1,2])` (replacing the term with a random asserted parent at each step) to model imprecise observation; (iii) append 3 in-vocabulary noise terms drawn from the indexed pool (`rng.sample(noise_pool, 3)`). The resulting query carries no gold-entity identifier, only term ids, rendered to text via the identical term-name lookup used at indexing.

Indexing, splits, and seeds. Per seed, the eligible pool is shuffled and truncated to `n_index`= 2000; a `holdout_frac`= 0.1 slice is reserved as held-out (NOVEL) entities (their true key is never indexed and feeds the abstain/novelty metrics), the remainder is indexed. The rare/long-tail slice is defined by information content (rarest gold term above the corpus median), computed over the *indexed* records only (no test-on-train leak). Counts are pooled over 3 seeds $\{7, 17, 23\}$ with 120 queries per seed; reported $n_{\text{all}}/n_{\text{rare}}/n_{\text{novel}}$ are these pooled totals (cyber is smaller because few intrusion-sets meet the 7–30 band).

Fairness (identical-record protocol). Every memory (SMA, BM25, BGE-dense, hybrid-RRF, hybrid+rerank, HippoRAG) indexes the *byte-identical* flattened term-name record built once per entity (`IndexItem.text = " ".join(term_name(t))`); only the retrieval mechanism differs. Crucially, is-a ancestors are *deliberately withheld* from the baseline documents and queries: SMA reaches ancestors at match time through bounded lattice ascension, whereas adding the ancestor closure to the baseline text was tested directly and *hurts* the baselines (common ancestors dilute IDF and cosine; Extended Data Table 4), so withholding them is the fair, not the favourable, choice.

Cyber alias guard. The mounted ATT&CK graph contains *only* technique terms (keyed by external id, e.g. `T1059.001`) and tactic terms; `intrusion-set` objects (threat-group names) and their `x_mitre_aliases` are never materialized as terms (`sma/ontology/attack.py`). Group names are used solely as gold entity keys, so no threat-group name or alias can enter any indexed document or query text, and the attribution task therefore cannot be solved by name leakage.

†Legal: CPC's near-uniform information content yields no rare split ($n_{\text{rare}} = 0$ of 324); legal is scored on the all-query slice. ‡We redistribute derived gold labels (disease ids and phenotype sets), not OMIM source records.

Extended Data Table 7 | Licensing of mounted ontologies and benchmark gold sources. Every mounted ontology and every derived gold-label source is openly licensed and version-pinned. We redistribute only *derived* gold (disease identifiers and phenotype/annotation sets) and never source records from any vocabulary that does not permit it; the OMIM caveat is noted below. Ontology release versions and file `md5` checksums are recorded in `data/manifests/datasets.json`.

Source	**Role**	**License**	**Redistributed**
HPO [14]	Ontology (medicine)	CC BY 4.0	ontology
GO [16]	Ontology (genomics)	CC BY 4.0	ontology
MONDO [15]	Ontology (disease)	CC BY 4.0	ontology
Uberon	Ontology (anatomy)	CC BY 4.0	ontology
ChEBI	Ontology (chemistry)	CC BY 4.0	ontology
Orphanet	Gold annotations (medicine)	CC BY 4.0	derived gold
GO Annotation File (GOA)	Gold annotations (genomics)	CC BY 4.0	derived gold
MITRE ATT&CK [18]	Ontology + gold (cyber)	Apache-2.0	ontology + gold
MITRE CAPEC / CWE	Ontology (cyber)	public terms of use	ontology
CPC	Ontology + gold (legal/IP)	public domain	ontology + gold
US-GAAP	Ontology (finance)	public domain	ontology
FIBO	Ontology (finance)	MIT	ontology
LKIF-core	Ontology (legal)	open (LGPL-compatible)	ontology
SEC EDGAR filings	Gold (finance)	public domain	derived gold
USPTO patent CPC assignments	Gold (legal/IP)	public domain	derived gold
OMIM	Gold annotations (medicine)	restricted	**derived only**†

†**OMIM caveat.** We do not redistribute OMIM source records. For the medicine arm we release only *derived* gold labels (disease identifiers paired with their phenotype (HPO term) sets, obtained through the openly licensed HPO disease–phenotype annotation file), and never OMIM clinical-synopsis text or any OMIM source field. Users requiring OMIM source records must obtain them directly from OMIM under its own terms. SNOMED CT and UMLS are deliberately excluded in favour of fully open reproducibility (see Limitations).

Extended Data Table 8 | SMA versus *every* retrieval baseline, with selection-corrected significance. Tail top-5 (rare slice; legal on the all-query slice) advantage $\Delta = \text{SMA} - \text{baseline}$ for each of the five baselines in each domain; all $\Delta > 0$. The right block reports the data-selected "best RAG" comparator (chosen by all-query top-5), its raw paired-bootstrap p, and the Bonferroni-over-baselines ($\times 5$) corrected p_{Bonf}. Source: `reports/confirmatory/agentic_vs_all_baselines.csv`.

	Δ tail top-5 vs. each baseline					vs. best (selected)		
Domain	BM25	Dense	Hyb-RRF	Hyb+Rr	HippoRAG	p_{raw}	p_{Bonf}	Survives
Medicine	+0.46	+0.45	+0.44	+0.34	+0.59	0.0002	0.0010	yes
Genomics	+0.30	+0.17	+0.15	+0.20	+0.53	0.0002	0.0010	yes
Finance	+0.17	+0.23	+0.19	+0.22	+0.32	0.0002	0.0010	yes
Legal†	+0.27	+0.07	+0.15	+0.15	+0.52	0.0022	0.0110	yes
Cyber	+0.06	+0.14	+0.02	+0.11	+0.39	0.0346	0.1730	**no**

Δ values are the marginal SMA tail top-5 (rare slice) minus each baseline's tail top-5; legal† uses the all-query slice (CPC has no rare split). These per-baseline marginal differences are distinct from the per-query paired-bootstrap mean Δ reported against the committed best baseline in Table 2, and for the narrow cyber margin they are smaller than it. The "vs. best" columns correct the data-selected comparator by Bonferroni over the five baselines. Four of five domains survive ($p_{\text{Bonf}} \leq 0.011$); cyber is directional ($p = 0.035$, $p_{\text{Bonf}} = 0.17$). $p_{\text{raw}} = 0.0002$ is the paired-bootstrap floor ($1/(R{+}1)$, $R = 10^4$). The bootstrap is per-query, not per-entity (three seeds share entities), so reported p-values are anti-conservative at the entity level.

Extended Data Table 9 | Mechanism ablation: what structure-mapping adds, one component at a time. Starting from a lexical-only (BM25) base, we add the is-a ascension lattice, then information-content (surprisal) weighting, then higher-order typed relations, and report rare-slice tail top-5 accuracy at each rung. Δ is the paired-bootstrap difference against the immediately preceding rung (3 seeds, n_{index}=400, n_{query}=90). Medicine (HPO) is a pure-is-a control with no typed relations to add; genomics (GO) carries typed relations. The is-a rung is the dominant lever (HPO +0.368, GO +0.080); IC weighting adds a significant second increment; typed higher-order relations add no measurable top-5 accuracy. All numbers from `reports/confirmatory/ablation_mechanism.csv`.

Domain	**Rung (cumulative mechanism)**	**Top-5 (rare)**	**Δ vs. prev**	**95% CI**	p
Genomics (GO)	lexical-only (BM25)	0.750	—	—	—
	+ is-a ascension	0.830	+0.080	[−0.005, +0.160]	0.066
	+ IC weighting	0.910	+0.080	[+0.047, +0.118]	$2{\times}10^{-4}$
	+ higher-order / typed	0.915	+0.005	[+0.000, +0.014]	0.72 (n.s.)
Medicine (HPO, pure-is-a control)	lexical-only (BM25)	0.587	—	—	—
	+ is-a ascension	0.955	+0.368	[+0.296, +0.440]	$2{\times}10^{-4}$
	+ IC weighting	0.982	+0.027	[+0.009, +0.049]	0.003
	+ higher-order / typed	0.982	+0.000	[+0.000, +0.000]	—

Rungs are cumulative: each row adds one mechanism to the row above. Δ is the paired-bootstrap difference of adjacent rungs (rare-slice tail top-5, top-5 tail). HPO is a pure-is-a ontology, so its +higher-order rung adds no typed relations and yields Δ=+0.000 by construction. The advantage decomposes into is-a subsumption (the dominant lever) plus information-content weighting (a significant second increment); typed higher-order relations add no measurable top-5 accuracy (GO +0.005, n.s.), consistent with the Phenomizer IC parity (Extended Data Table 4) and the SSB structure-only control (Extended Data Table 3). The ablation runs at a reduced index scale (n_{index}=400; the full-scale run is compute-bound), so the robust quantity is the *relative* contribution of each rung rather than the absolute accuracies.

Extended Data Table 10 | Entity-clustered robustness of the memory-swap result. The committed headline bootstrap resamples per query, but each target entity contributes up to three correlated queries (one per seed). We re-ran the identical evaluation with a *cluster bootstrap* that resamples entities with replacement (taking all of an entity's queries together); the per-query point estimate is reproduced exactly and then re-aggregated by entity. Shown are the per-query Δ and p and the entity-clustered Δ, 95% CI and p, for SMA versus the committed best baseline. Like the mechanism ablation, this re-analysis runs at a reduced index scale (n_{index}=250), so the Δ values here are smaller than the full-scale marginals in Table 2 and are not directly comparable to them; the robustness question is whether resampling by entity rather than by query changes the significance verdict. It does not for the strong domains, whose entity-clustered intervals exclude zero, but it does for cyber, whose clustered interval includes zero. Cyber was additionally verified at full scale (n_{index}=2,000): the clustered 95% CI is $[-0.008, +0.155]$, p=0.088, versus per-query p=0.035. Source: `reports/confirmatory/cluster_bootstrap.csv`.

Domain	Best baseline	Δ (per-query)	p (per-query)	Δ (clustered)	p (clustered, 95% CI)
Medicine	Hybrid+rerank	+0.178	2×10^{-4}	+0.178	2×10^{-4} $[+0.137, +0.220]$
Genomics	Dense	+0.053	0.007	+0.053	0.009 $[+0.014, +0.092]$
Finance	Hybrid-RRF	+0.225	2×10^{-4}	+0.225	2×10^{-4} $[+0.153, +0.300]$
Legal	Dense	+0.050	2×10^{-4}	+0.050	2×10^{-4} $[+0.028, +0.075]$
Cyber	Hybrid-RRF	+0.073	0.035	+0.073	0.088 $[-0.008, +0.155]$

Extended Data Table 11 | Computational cost: no embeddings, no GPU, no vector store. Index-build time, per-query latency (p_{50}/p_{95}) and peak resident memory for SMA, lexical BM25, and BGE-small dense retrieval, measured on the medicine arm (n_{index}=2,000; 200 queries) on a single CPU (AMD Ryzen 7 6800H, 16 threads), each arm timed in its own process. SMA requires no learned embedding model, no GPU and no vector store, and its memory footprint is $\approx 5.5\times$ smaller than dense retrieval; the trade-off it pays is a slower CPU-bound build and a higher per-query latency than BM25. We therefore claim zero neural-indexing cost, not lower latency. Source: `reports/confirmatory/cost_latency.csv`.

System	Embed. model	GPU	Vector store	Build (s)	p_{50} (ms)	p_{95} (ms)	RAM (MB)
SMA	none	no	no	566	241	360	250
BM25	none	no	no	<0.01	23	33	187
Dense (BGE-small)	`bge-small`	optional	yes	33	151	167	1,378